

Introduction

The Gambler's Ruin is a random walk, one of the most classical models of a Markovian stochastic process. In this model, a gambler wagers money in each round and, according to a probabilistic rule, either gains or loses varying amounts. The game ends when the gambler either reaches the target fortune or goes bankrupt.

While the probability of ruin has been extensively studied, the expected duration of the game remains less explored for more general game configurations. This project investigates methods to estimate this expected time by combining analytical tools with computational simulations.

Walk on $\mathbb{Z}^1$

Model's Definition

Player A starts with an initial fortune of $\$a$, while $\$N$ are in play. Each round produces an independent random outcome X_i , $i = 1, 2, \dots$, representing the gain or loss at that step. Henceforth, this will be called the "rule" variable. After n rounds, the fortune is

$$S_n = a + \sum_{i=1}^n X_i. \quad (1)$$

The **duration of the game** is the first time n such that $S_n \leq 0$ (ruin) or $S_n \geq N$ (victory) and is denoted as $\tau_{X,a,N}$.

This model can be extended for walks on random environments by indexing X on the walk's current position.

From the definition above, we consider the simplest case of the **Gambler's Ruin**, where, at each round, the player wins $\$1$ with probability p and loses $\$1$ with probability $q = 1 - p$. That is, X_i is defined such that $\mathbb{P}(X_i = +1) = p$ and $\mathbb{P}(X_i = -1) = q$.

It's possible to obtain the expected duration and variance of this random walk through conditional probability, as shown by Stern [1], and finite difference equations, as shown by Andel [2]. However, these methods are unsuitable to deal with non-trivial rule variables.

A First Estimator

Let S_n a random walk with rule variable X_i such that $\mathbb{E}(X_i) \neq 0$. We define the Estimator for the expected duration of such walks in $\mathbb{Z}^1$ as

$$\widehat{\mathbb{E}}[\tau_{X,a,N}] = \begin{cases} \frac{N-a}{\mathbb{E}[X_i]} & \text{if } \mathbb{E}[X_i] > 0, \\ \frac{a}{\mathbb{E}[X_i]} & \text{if } \mathbb{E}[X_i] < 0. \end{cases} \quad (2)$$

We arrived at Equation 2 through a heuristic examination of uniform linear motion (ULM) in Physics. We drew a parallel between velocity and the expected value of the rule variable, and between distance and fortune. Through Monte Carlo simulation, as recommended by Ritter [3], we compared the estimator and the simulated expected value. The results are shown in Figure 1 below.

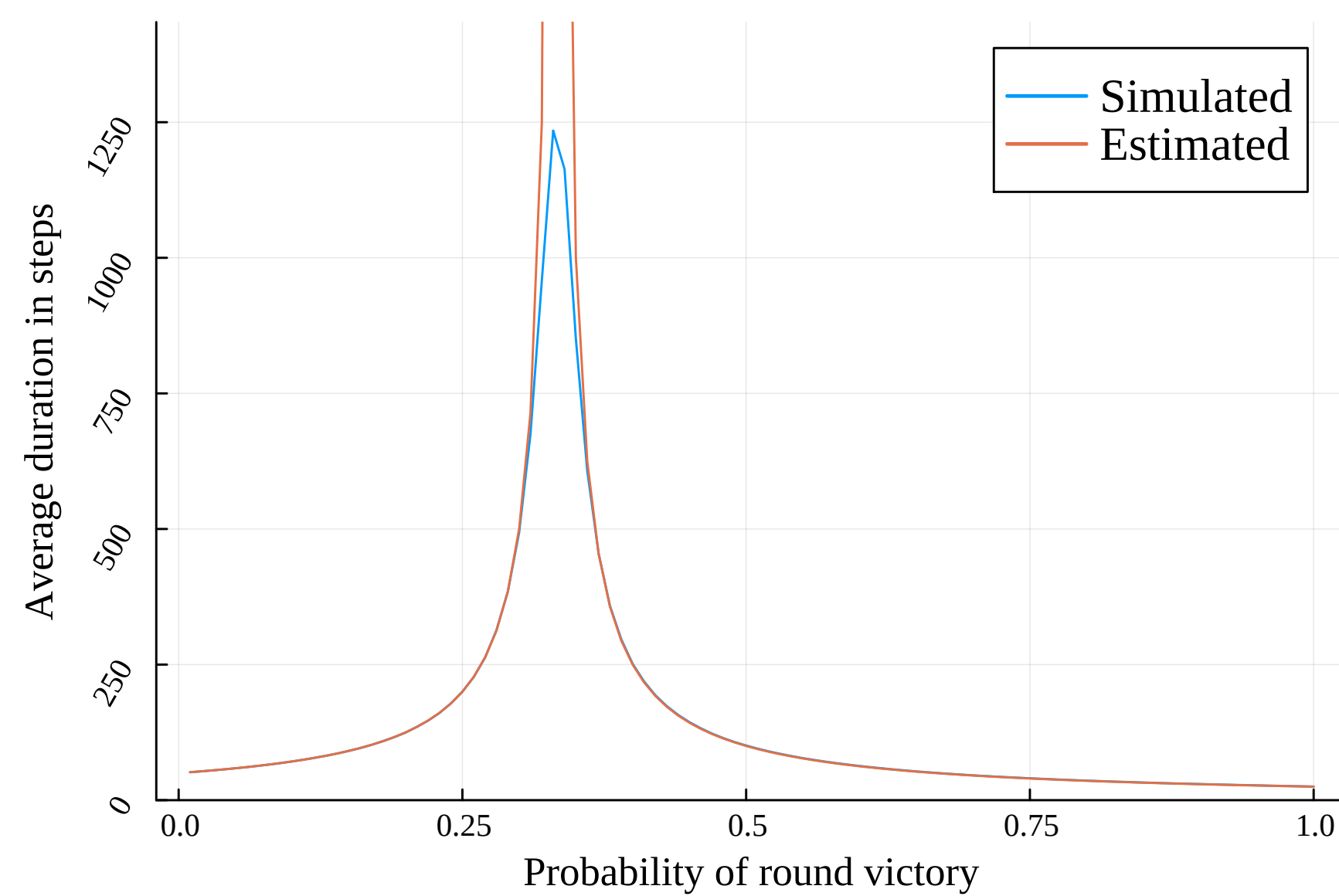


Figure 1: Estimator comparison with 100 000 simulations per probability of round victory

The graph in Figure 1 highlights the estimator's general precision for most values of p . One important disadvantage to note is its increasing error rate as $\mathbb{E}[X_i]$ approaches 0. In this initial form, the estimator described in Equation 2 is also unsuitable to handle games where no money might be exchanged in a round (a draw).

Estimating in Random Environments

Let $\mathbf{R} = (R_1, R_2, \dots, R_N)$ a random environment. From now on, the rule X_n is defined based on the current fortune S_n of the gambler. Let $C_i^- = \sum_{j=1}^i \frac{1}{R_j}$ and $C_i^+ = \sum_{j=i}^N \frac{1}{R_j}$. We define the Estimator for the expected duration of such walks in $\mathbb{Z}^d$ as

$$\mathbb{E}[T_{\mathbf{R},a,N}] = \begin{cases} C_i^- & \text{if } \mathbb{E}[R_a] < 0 \\ C_i^+ & \text{if } \mathbb{E}[R_a] > 0 \end{cases} \quad (3)$$

This generalizes the estimator defined in Equation 2 to most random environments.

The estimator was shown to be satisfactory as depicted in Figure 2. This is useful for providing computationally light estimations for the duration of such walks.

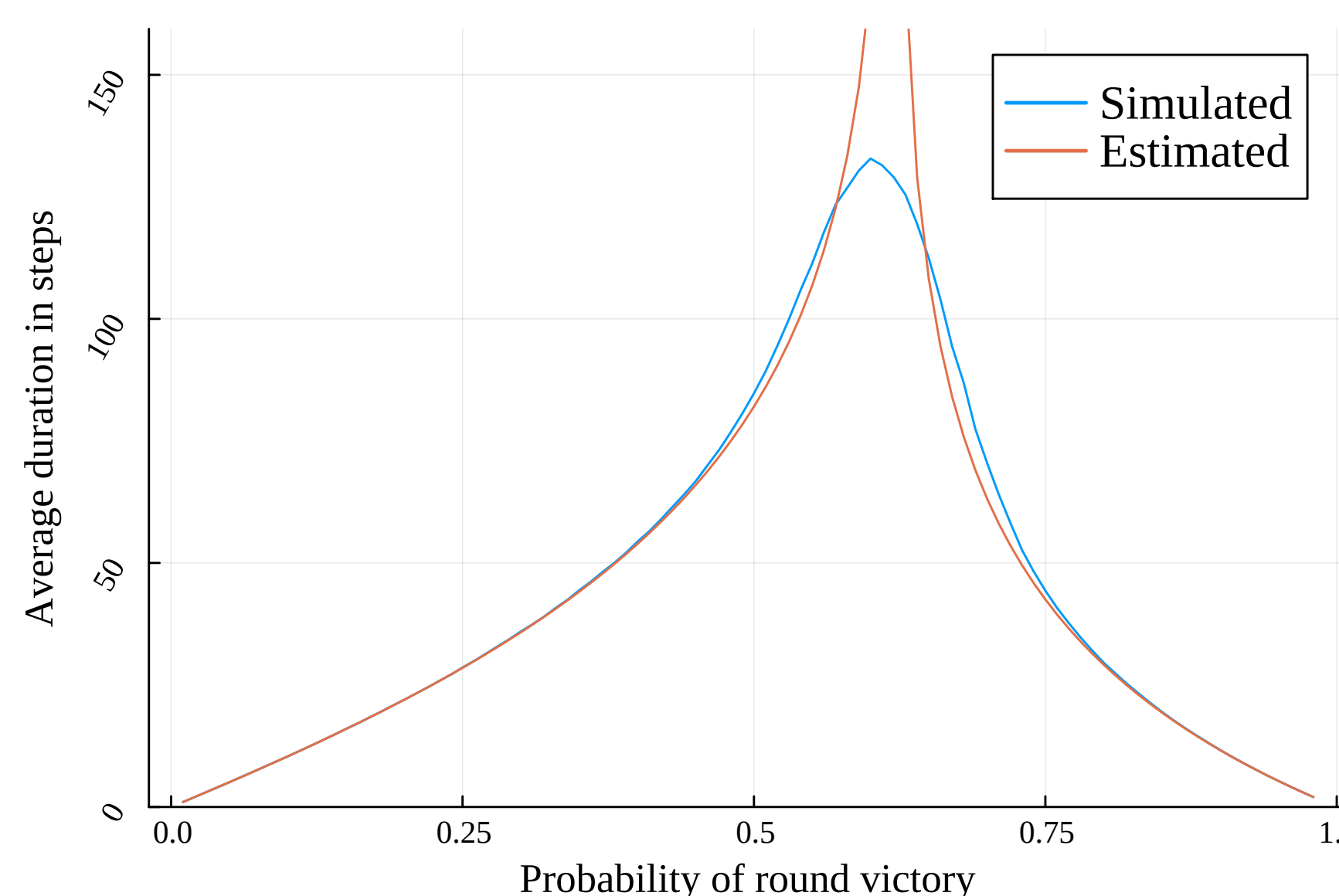


Figure 2: Comparison of estimative in random environment with 100 000 simulations per probability of round victory

Similar unsuitableness as seen in the first estimator applies, such as when $\mathbb{E}[X_i] \approx 0$ for any $i = 1, \dots, N$.

On top of that, we've observed that this estimator deviates considerably from the simulated values in some environments.

Walk on $\mathbb{Z}^d$

Model's Definition in higher dimensions

Player A starts with an initial fortune vector $\mathbf{a} = (a_1, a_2, \dots, a_d)$, while $\mathbf{N} = (N_1, N_2, \dots, N_d)$ are in play, where each N_i , $i = 1, 2, \dots, d$ represents the available fortune for that *sub-game*. Each round produces an independent random vector outcome $\mathbf{X}_i$ of the form $(X_{i,1}, X_{i,2}, \dots, X_{i,d})$, $i = 1, 2, \dots$, representing the gain or loss at that step, the "rule" variable. After n rounds, the fortune is

$$\mathbf{S}_n = \mathbf{a} + \sum_{i=1}^n \mathbf{X}_i. \quad (4)$$

The **duration of the game** is the first instant n such that there exists an element W of $\mathbf{S}_n$, such that $W \leq 0$ (ruin) or $W \geq N_i$ (victory) and is denoted as $\tau_{\mathbf{X},\mathbf{a},\mathbf{N}}$.

This model can be extended for walks on random environments by indexing X on the walk's current position.

For $\mathbb{Z}^2$, we used computational symbolic systems solvers to evaluate the expected game's duration for different $\mathbf{a}$. This method works for every $\mathbb{Z}^d$, $d \in \mathbb{Z}^+$, but the computational complexity for solving the system grows exponentially as d or the state space increases. The result of our calculations in $\mathbb{Z}^2$ can be seen in the heatmap from Figure 3, which resulted from random walks with rule $\mathbf{X}$ such that $\mathbb{P}(\mathbf{X}_i = (1, 0)) = 0.3$, $\mathbb{P}(\mathbf{X}_i = (-1, 0)) = 0.2$, $\mathbb{P}(\mathbf{X}_i = (0, 1)) = 0.25$, $\mathbb{P}(\mathbf{X}_i = (0, -1)) = 0.25$, and $\mathbf{N} = (15, 15)$.

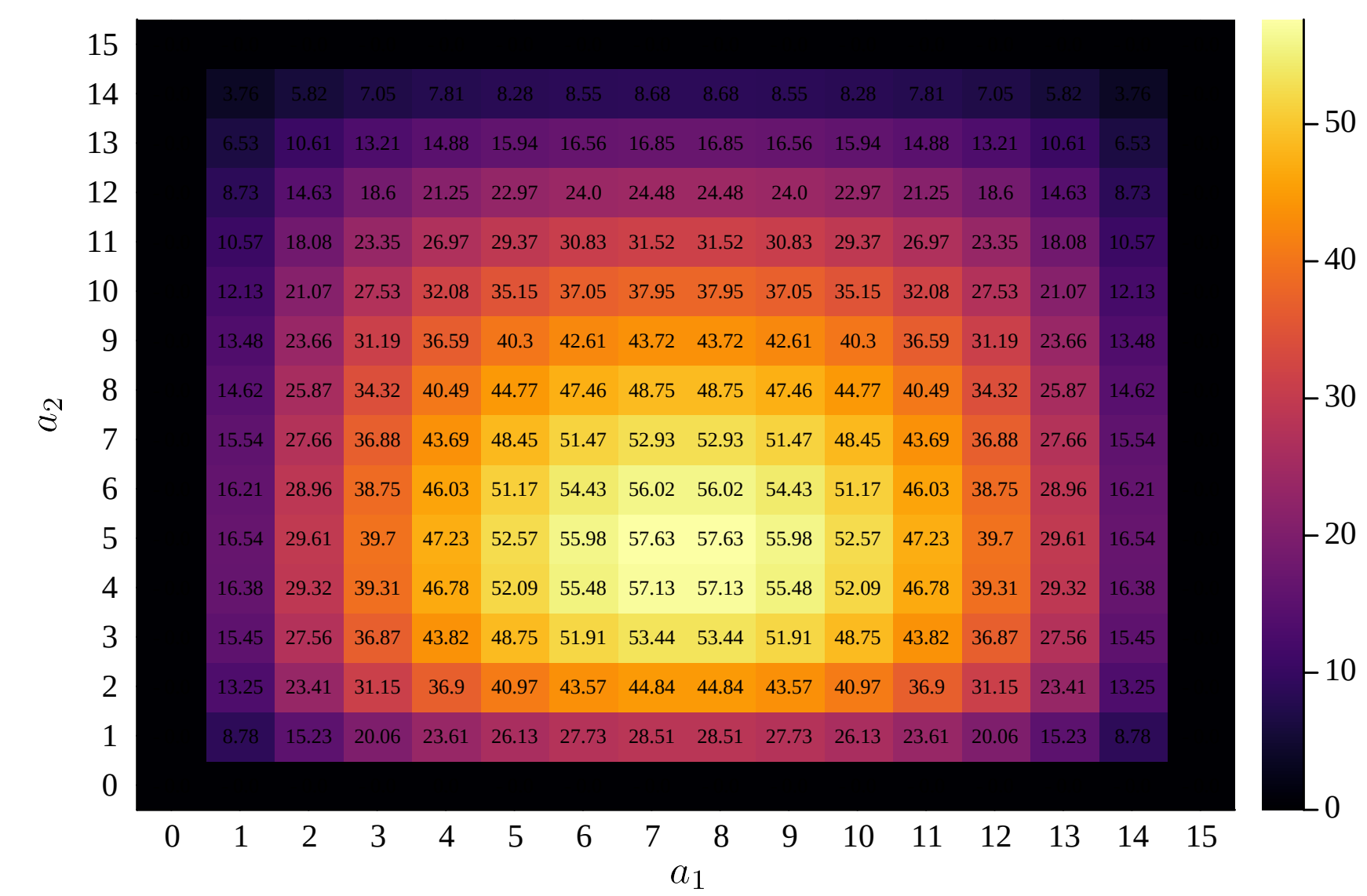


Figure 3: Expected game duration in $\mathbb{Z}^2$ as a function of the initial fortune (a_1, a_2) .

In higher dimensions, we define a game which rules depend on d in such a way that it remains unitary and fair. That is, the expected profit in coordinate is always zero, exactly $\$1$ is wagered in every round, and the player bets with equal probability for every coordinate. As the dimensionality increases, the average duration of the game grows approximately linearly, as illustrated in Figure 4.

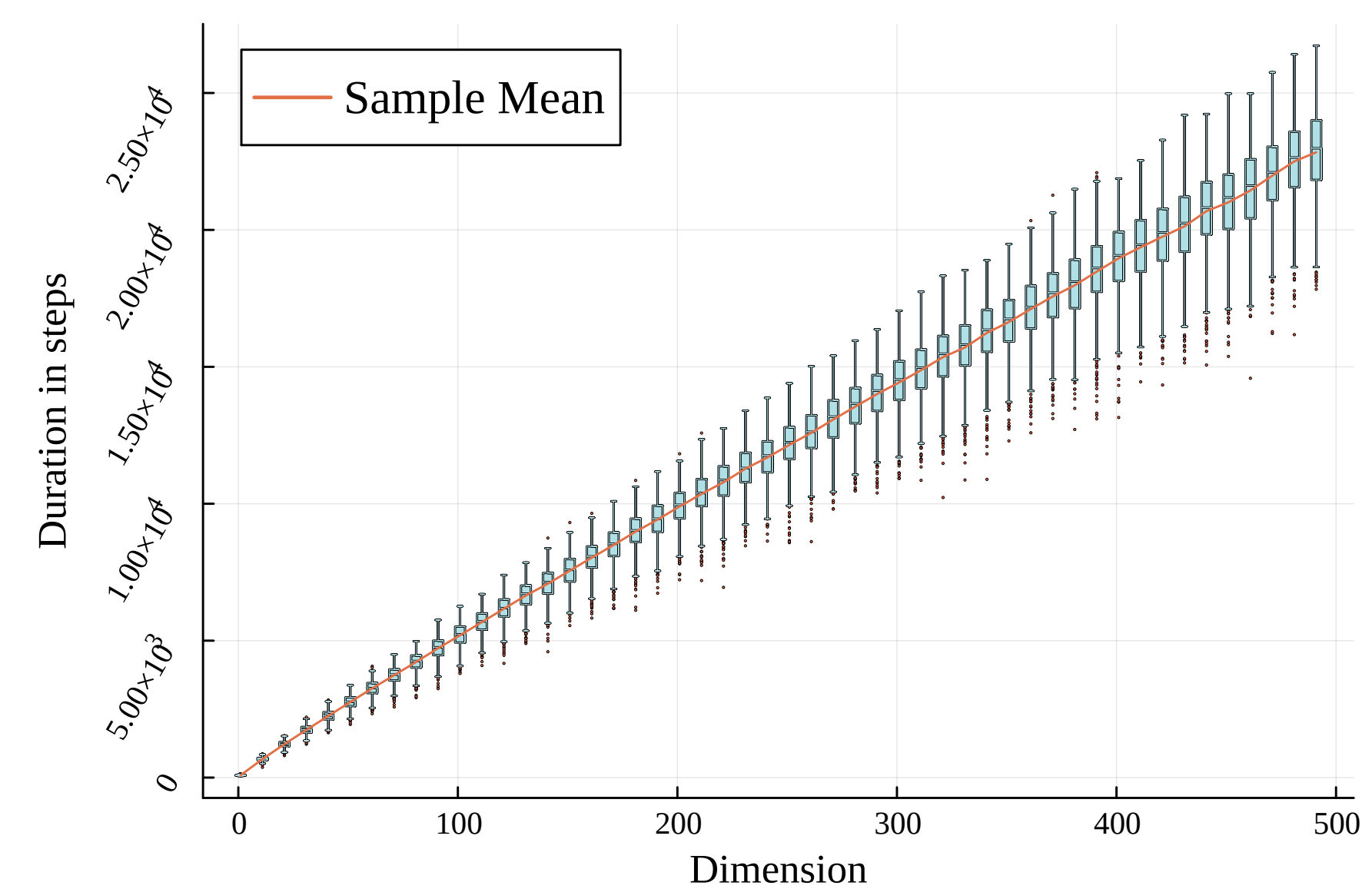


Figure 4: Boxplot series of 1 000 simulated durations per Dimension

Conclusions

- A **heuristic estimator** for the expected duration in $\mathbb{Z}^1$ was developed, based on an analogy with uniform linear motion, showing good accuracy in non-critical regimes.
- The estimator was extended to **random environments**, providing computationally efficient approximations where analytical solutions are not feasible.
- The model was **generalized to higher dimensions**, allowing for analytical results through computational means.
- Numerical experiments indicate that the **expected duration grows approximately linearly as does the dimensionality**, revealing a robust structural property of such random walks.
- **Computational bottlenecks** are significant: the computational complexity for exact or symbolic calculation grows rapidly with the number of states and dimensionality, making simulations increasingly costly. This issue is mitigated, but still persists, when adopting high-performance programming languages, such as Julia [4], and multi-threading. Computer memory was an even bigger problem in solving such systems, highlighting computational challenges in both RAM and processing usage.

References

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- [4] W. F. Godoy *et al.*, "Evaluating performance and portability of high-level programming models," May 2023.

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